Team Vidhidnya

Google Developer Groups

BlockVerse 2K26

Introduction

Team Name :- Team Vidhidnya (विधिज्ञ)

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College Name :- K. K. Wagh Institute Of Engineering Education & Reasearch

Domain / Track :- AI/ML in Legal Tech

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Problem Statement

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Title : LexisGraph: A Graph-Based Retrieval-Augmented Generation (GraphRAG) System for Automated Regulatory Gap Analysis.

Problem Description :

- Corporate Organizations struggle to maintain compliance with large volumes of complex and continuously evolving regulations.
- Existing approaches rely on manual audits and keyword-based systems that fail to capture relationships and context within legal data.
- As a result, critical dependencies, cross-references, and conflicts are missed, leading to incomplete compliance analysis, regulatory violations, and significant legal risk.

Target Users / Stakeholders :

Stakeholder	Pain Point	Lexisgraph
Compliance Officers	Struggling to deal with notifications from laws on a daily basis.	Automates the monitoring and gap analysis.
Legal Teams	Costly process of manual cross-referencing of documents.	Creates a legal relationship graph.
Startups	Budget constraints for huge legal departments.	Provides an “AI-Auditor” to achieve day one compliance.
External Auditors	External Auditors: Challenge of validating continuous compliance.	Creates graph based audit trails.

Significance of the Problem :

- **High Risk of Non-Compliance:** Undetected gaps and conflicts increase the likelihood of regulatory violations and legal penalties.
- **Operational Inefficiency:** Manual compliance audits are time-consuming, resource-intensive, and difficult to scale.
- **Inability to Handle Regulatory Changes:** Traditional systems fail to keep up with continuously evolving laws and regulatory updates.
- **Limited Visibility and Explainability:** Lack of clear insights makes it difficult for organizations to understand and justify compliance decisions.

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LexisGraph

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Proposed Solution :

- LexisGraph is an AI-based platform that analyzes **internal policies** against **regulatory documents** to detect compliance gaps, and conflicts.
- The system transforms **unstructured legal text** into a structured **knowledge graph**, representing clauses as **interconnected entities** with relationships such as dependencies, conflicts, and amendments.
- Using a Graph-Based Retrieval-Augmented Generation (**GraphRAG**) approach, it **retrieves contextually** relevant legal information through multi-hop reasoning across the knowledge graph.
- This enables deep **semantic understanding** and **accurate mapping** between regulatory requirements and policies in complex scenarios.
- Based on this analysis, the **system identifies** compliance gaps, including missing, conflicting, and outdated clauses within **policies**.
- The system generates clear, detailed, and explainable **compliance reports** with traceable legal references, enabling organizations to maintain alignment with **continuously evolving regulations**.

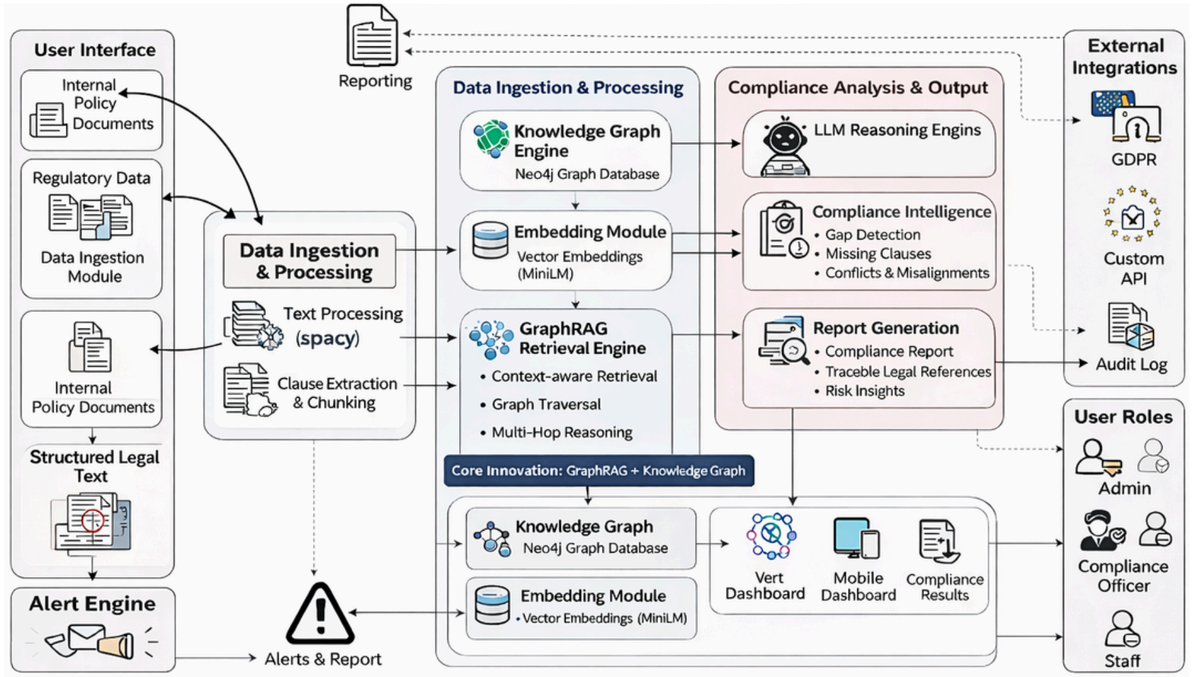
Key Features & Uniqueness :

- **Knowledge Graph Engine:** AI-powered engine ingests regulatory and policy data and structures it into a unified knowledge graph with clause-level links.
- **Retrieval Engine:** GraphRAG engine performs context-aware retrieval with multi-hop reasoning, going beyond traditional keyword-based approaches.
- **Relational Reasoning:** Reasoning engine maps regulations with internal policies to identify relationships and inconsistencies in legal structures.
- **Compliance Intelligence:** Compliance engine detects missing, conflicting, and outdated clauses and generates clear, explainable reports with traceable legal references for better compliance decisions.

Addressing the Problem :

- **Automation:** The system automates compliance analysis, replacing slow and error-prone manual auditing with a faster, scalable, and more reliable approach.
- **Semantic Understanding:** It overcomes the limitations of keyword-based and rule-driven systems by enabling accurate understanding of the context and meaning of legal documents.
- **Knowledge Graph:** By representing regulations and internal policies in a unified knowledge graph, it creates direct and structured mappings between legal requirements and organizational policies.
- **GraphRAG Reasoning:** Using GraphRAG and multi-hop reasoning, the system identifies hidden dependencies, cross-references, and conflicts that are typically missed by traditional methods.
- **Explainability:** It generates clear, explainable, and traceable compliance insights while continuously adapting to evolving regulations and legal updates.

System Flow



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Technical Approach

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System Architecture / Workflow

Layer 1: Data Layer

User Input
Policies (PDF)

Realtime Data
Regulatory Sources (Web)

→ **Preprocessing**
(Clause Extraction)

Layer 2: Intelligence Layer

Knowledge Graph
Nodes = Clauses
Edges = Relations

Embeddings
Vector Representation

→ **GraphRAG Retrieval**
Retrieval Logic

Layer 3: Reasoning Layer

1. LLM Analysis
2. Gap Detection
3. Compliance Report

Flowchart

LexisGraph – User Journey
From Upload to Compliance Insights

1. Sign Up / Login
Create your account or securely login to access your compliance workspace.

2. Upload Internal Policies
Upload your organization's policy documents in PDF or DOC format.

3. Select Regulations (Optional)
Choose relevant laws or regulatory frameworks for analysis (preloaded or custom).

4. Start Compliance Analysis
Initiate the analysis with a single click. LexisGraph reviews policies against regulations.

5. Review Compliance Results
View detected gaps, conflicts, and outdated clauses with clear explanations and references.

6. Explore Insights
Analyze risk areas and visualize key findings through easy-to-understand summaries and charts.

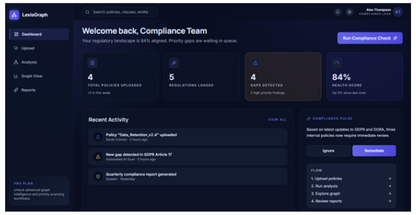
7. Download Reports
Download detailed compliance reports for review, sharing, or audit purposes.

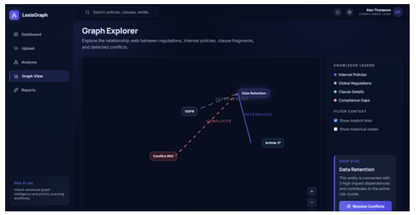
8. Take Action
Update policies, close gaps, and strengthen compliance with confidence.

Smarter Compliance. Faster Decisions. Greater Confidence.

Live Demo :- https://lexis-graph.vercel.app/ (Click)

Prototype :





Technology Stack:

Website
React JS

AI/ML
LLM

Database
neo4j
PostgreSQL

Tools, APIs, and Frameworks Used :

- **NLP:** spaCy (Entity Extraction)
- **Graph:** Neo4j + Graph algorithms
- **RAG Pipeline:** LangChain
- **Frontend UI:** React + Chart.js / React Flow
- **Deployment:** Vercel (Frontend)
- **Version Control:** Git + GitHub

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Feasibility and Viability

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Technical Feasibility :

- LexisGraph is technically feasible using existing and well-established technologies such as NLP, Knowledge Graphs, and Large Language Models.
- The system leverages NLP techniques to process unstructured legal documents and extract meaningful clauses and relationships.
- A graph database (Neo4j) enables efficient representation of legal structures and supports relationship-aware analysis through knowledge graphs.
- GraphRAG integrates knowledge graphs with language models to enable context-aware retrieval and multi-hop reasoning across complex legal data.
- The system can be implemented using scalable technologies such as Python (FastAPI), React, and modern AI frameworks, ensuring reliable and efficient performance.

Resource & Cost Considerations :

- The system primarily relies on open-source technologies such as Python, Neo4j, and NLP libraries, reducing initial development costs.
- Cloud infrastructure or local servers can be used for deployment, allowing flexible cost management based on usage and scale.
- Large Language Models can be accessed through APIs or lightweight models can be deployed locally to optimize operational costs.
- Storage and processing requirements can be managed efficiently using scalable databases and modular system architecture.

Scalability and Sustainability :

- The system can efficiently handle large volumes of regulatory and policy documents through graph-based storage and retrieval mechanisms.
- Its architecture supports incremental updates, allowing new regulations and policy changes to be integrated without rebuilding the entire system.
- Graph-based modeling enables scalable handling of complex legal relationships such as cross-references, dependencies, and amendments.
- The system can be extended across multiple domains and industries, supporting cross-regulation compliance analysis.
- Use of modular components allows independent scaling of data ingestion, graph processing, and reasoning layers.
- Integration with cloud infrastructure enables horizontal scaling based on workload and user demand.
- The platform supports continuous compliance monitoring, ensuring long-term adaptability to evolving regulatory environments.
- The system supports distributed processing for data ingestion and analysis, enabling faster handling of large-scale regulatory datasets.
- The knowledge graph structure minimizes redundant computations by reusing relationships, improving efficiency as the system scales.

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Impact and Benefits

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Impacts :

Enterprises / Organizations :

LexisGraph enhances organizational compliance workflows by automating policy analysis and significantly reducing reliance on manual auditing processes, leading to improved operational efficiency and consistency.

Compliance & Legal Teams :

Enables faster, more accurate, and reliable compliance verification by identifying legal gaps and risks early, helping organizations avoid penalties and ensure regulatory adherence.

Decision Makers / Management :

The system provides transparent and explainable compliance insights, enabling more informed and data-driven decision-making through clear visibility into policy gaps, risks, and regulatory alignment.

Regulatory & Dynamic Environments :

LexisGraph supports continuous compliance by adapting to frequently changing regulations, ensuring organizations remain up-to-date in dynamic and complex legal environments.

Complex Legal Understanding :

LexisGraph enables organizations to understand complex legal structures, including cross-references, exceptions, and dependencies often missed in traditional methods.

Benefits :

- **Automated Comparison:** LexisGraph allows organizations to automatically compare their internal policies with relevant regulations, significantly reducing the time and effort required for compliance audits.
- **Gap Detection:** It helps identify missing, conflicting, and outdated clauses within policies, improving the accuracy and reliability of compliance verification.
- **Structured Understanding:** The system simplifies complex legal documents by structuring them into a connected knowledge graph, making them easier to understand and analyze.
- **Explainable Insights:** It generates clear and explainable compliance reports with proper legal references, enabling users to understand not just what the issue is, but why it exists.
- **Continuous Compliance:** LexisGraph supports continuous compliance by allowing organizations to quickly re-evaluate policies as regulations change over time.

Future Scope and Enhancements :

- The system can allow organizations to **compare their policies with industry standards** and **best practices** to understand where they stand.
- It can include a feature where users can **test policy changes** and see their **impact on compliance** before applying them in **real scenarios**.
- The platform can **track** changes in policies at the **clause level**, helping users understand how **policies evolve over time** with **regulatory updates**.
- The system can maintain a **history of compliance** status, allowing organizations to **review past performance** and **track improvements** over time.